

2.1 Foundations in biology

2.1.1 Cell structure

Biology is the study of living organisms. Every living organism is made up of one or more cells, therefore understanding the structure and function of the cell is a fundamental concept in the study of biology. Since Robert Hooke coined the phrase 'cells' in 1665, careful

observation using microscopes has revealed details of cell structure and ultrastructure and provided evidence to support hypotheses regarding the roles of cells and their organelles.

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) the use of microscopy to observe and investigate different types of cell and cell structure in a range of eukaryotic organisms	To include an appreciation of the images produced by a range of microscopes; light microscope, transmission electron microscope and scanning electron microscope. HSW1, HSW7
(b) the preparation and examination of microscope slides for use in light microscopy	Including the use of an eye piece graticule and stage micrometer. PAG1 HSW4
(c) the use of staining in light microscopy	To include the use of differential staining to identify different cellular components and cell types. PAG1 HSW4, HSW5
(d) the representation of cell structure as seen under the light microscope using drawings and annotated diagrams of whole cells or cells in sections of tissue	PAG1

(e) the use and manipulation of the magnification formula	$\text{magnification} = \frac{\text{size of image}}{\text{size of real object}}$
(f) the difference between magnification and resolution	<i>M0.1, M0.2, M0.3, M1.1, M1.8, M2.2, M2.3, M2.4</i> To include an appreciation of the differences in resolution and magnification that can be achieved by a light microscope, a transmission electron microscope and a scanning electron microscope. Learners are not required to recall exact resolutions or magnification numbers for each of the listed microscopes.
(g) the ultrastructure of eukaryotic cells and the functions of the different cellular components	<i>M0.2, M0.3</i> HSW7, HSW8 To include the following cellular components and an outline of their functions: nucleus, nucleolus, nuclear envelope, rough and smooth endoplasmic reticulum (ER), Golgi apparatus, ribosomes, mitochondria, lysosomes, chloroplasts, plasma membrane, centrioles, cell wall, flagella and cilia.
(h) photomicrographs of cellular components in a range of eukaryotic cells	<i>M0.2</i> To include interpretation of transmission and scanning electron microscope images.
(i) the interrelationship between the organelles involved in the production and secretion of proteins	To include providing mechanical strength to cells, aiding transport within cells and enabling cell movement.
(j) the importance of the cytoskeleton	HSW2
(k) the similarities and differences in the structure and ultrastructure of prokaryotic and eukaryotic cells.	PAG1